Project Design Phase-I
Solution Architecture

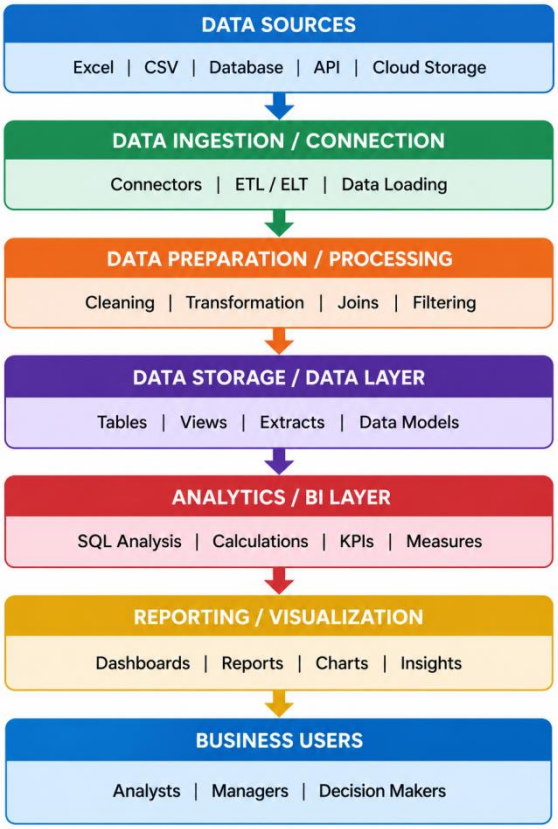
Date	20 October 2026
Team ID	En25cs3010891
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Collects raw physiological metrics, EHR records, and external data feeds (Excel, CSV, Database, API, Cloud Storage).	IoT Wearables, Medical Sensors, APIs, Cloud Storage
Data Ingestion / Connection	Handles secure data streaming, connectors, ETL/ELT workflows, and data loading.	MQTT, Apache Kafka, REST APIs
Data Preparation / Processing	Cleans, transforms, joins, and filters incoming vital signs and patient data.	Python (Pandas, NumPy), Apache Spark
Data Storage / Data Layer	Organizes and stores structured and unstructured records using tables, views, extracts, and data models.	PostgreSQL, MongoDB, AWS S3
Analytics / BI Layer	Performs SQL analysis, calculations, health metrics tracking, KPIs, and machine learning measures.	Python (Scikit-Learn/TensorFlow), SQL
Reporting / Visualization	Delivers real-time dashboards, medical reports, charts, and actionable clinical insights.	React, Chart.js, HTML5/CSS
Business Users	End users including doctors, hospital managers, administrators, and patients who utilize the insights.	Web & Mobile Dashboards